Smart District Analytics

A Data-Driven Housing Selection Platform for Almaty

Technical Research Report

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Almaty, Kazakhstan

Abstract

This report presents the design, development, and evaluation of Smart District Analytics, an integrated data-driven platform for housing selection in Almaty, Kazakhstan. Urban residents face significant challenges when choosing where to live, as critical information about safety, air quality, housing prices, and public infrastructure is dispersed across multiple disjointed sources. This project consolidates those data streams into a single, interactive Streamlit application that leverages machine learning, time-series forecasting, and geospatial analytics to support evidence-based decision-making.

The platform integrates five heterogeneous datasets covering road traffic accidents (2015–2025), air pollution metrics (PM2.5/PM10), real estate listings scraped from Krisha.kz, and infrastructure location data for schools, hospitals, and universities across Almaty's eight administrative districts. A Random Forest classifier is trained to predict accident severity, a SARIMA model forecasts future accident volumes, and a regression model estimates housing price fairness. An unsupervised clustering component groups accidents by fatality profiles. The final application provides six interactive pages — Air Quality, Accidents, Housing, Forecast, Infrastructure, and District Score — alongside a novel Listing Analyzer feature that evaluates individual properties using a composite scoring system incorporating safety, ecology, pricing, and nearby services.

Iterative improvements across two beta versions significantly enhanced system reliability, code maintainability, input validation, and response speed. The report discusses the architecture, methodology, results, limitations, and directions for future research.

1. Introduction

The rapid urbanization of Central Asian cities has made the task of choosing a place of residence increasingly complex and data-intensive. Almaty, the financial and cultural capital of Kazakhstan, is home to more than two million residents and continues to attract significant internal and international migration. As a result, its residential real estate market is highly dynamic, with prices, safety conditions, environmental quality, and infrastructure accessibility varying considerably across its eight administrative districts: Almalinsky, Bostandyksky, Zhetysusky, Medeusky, Nauryzbaysky, Turksibsky, Alatausky, and Almaty District.

Despite the abundance of data generated by city agencies, environmental monitoring stations, road safety authorities, and property listing portals, this information has historically been fragmented and inaccessible to ordinary residents. Someone wishing to evaluate a specific apartment must manually consult government accident statistics, read separate ecological monitoring reports, browse real estate portals for price benchmarks, and independently verify proximity to schools or hospitals. This fragmentation imposes a high cognitive burden and routinely leads to suboptimal housing decisions — both financial and lifestyle-related.

Smart District Analytics was designed to address this gap. The platform brings together multiple public and scraped data sources into a unified analytical environment. By applying supervised and unsupervised machine learning, time-series forecasting, and geospatial proximity analysis, it transforms raw data into actionable district-level and listing-level insights. Users can compare districts on a multi-criteria scoring dashboard, explore air quality trends, examine historical accident patterns, browse filtered real estate listings, and analyze a specific property through an automated Listing Analyzer.

The project proceeded through two development phases. The first phase (Beta 1) focused on building the core data pipeline and ML models. The second phase (Beta 2 / Final) concentrated on system architecture improvements, input validation, caching, geolocation features, and user experience refinements. This report documents the complete project lifecycle, from dataset curation to deployment.

The primary contributions of this work are:

- A unified, open-source decision-support platform integrating safety, environmental, housing, and infrastructure data for Almaty.
- A machine learning pipeline featuring Random Forest-based accident severity prediction, K-Means clustering for fatality profiling, SARIMA accident forecasting, and regression-based housing price estimation.
- A novel Listing Analyzer that scores individual real estate properties across four weighted dimensions: price fairness, safety index, environmental quality, and nearby services.
- A geolocation-based proximity system that counts educational institutions, hospitals, and universities within configurable radii (500 m, 1,000 m, 2,000 m) from any property.
- An iterative software engineering effort that centralized configuration, introduced comprehensive input validation, and achieved a tenfold improvement in data loading speed through persistent caching.

2. Datasets

The platform is built on five primary datasets, each targeting a distinct analytical domain. Data collection combined official government records, scraped real estate listings, and environmental sensor outputs. The following subsections describe each dataset, its origin, scope, and preprocessing steps.

2.1 Road Traffic Accidents Dataset

The road traffic accident dataset (processed_table.csv) was sourced from the publicly available ArcGIS web map service operated by the Kazakhstan Road Police (KGP) at gis.kgp.kz. Since direct API access required institutional credentials, the dataset was retrieved through a professional channel: a former supervisor from an internship at the North Caspian Operating Company (NCOC) — an organization with established ArcGIS access — assisted in obtaining the raw data. This ensured the data was official, complete, and structurally consistent with the original source. The

dataset contains records of road traffic incidents in Almaty spanning the decade from 2015 to 2025. Each record includes the date and time of the accident, the type of road and intersection, weather conditions, lighting conditions, the number of vehicles involved, and a severity classification indicating whether the incident resulted in fatalities, serious injuries, or was a non-injury collision. After preprocessing, the dataset contains features suitable for both classification (severity prediction) and time-series analysis (accident volume forecasting). Raw data was cleaned to remove duplicate entries, impute missing coordinate fields using district-level centroid values, and encode categorical variables using label encoding. This dataset is the primary source for the Accidents and Forecast pages.

2.1.1 Associated Research and Capstone Integration

The above work underpins a research paper titled *"Data-Driven Road Accident Probability Modeling and Fatality-Based Clustering in Almaty: A Machine Learning Approach to Public Safety Improvement,"* which presents the full pipeline and its policy implications. Key findings identify foggy weather, drug intoxication, and absent night-time street lighting as the most critical compounding risk factors for fatal outcomes. Within the Smart District Analytics capstone, district-level accident frequency, severity rates, and cluster profiles form the safety dimension of the multi-factor liveability index, integrated alongside housing prices and environmental and infrastructure data to enable genuinely informed residential decision-making.

2.2 Air Quality Dataset

Air quality data is available in two forms: a raw station-level file (almaty_air_districts.csv) and a processed district-level aggregate (processed_air_ala_data.csv). Both contain measurements of particulate matter concentrations — specifically PM2.5 (fine particles smaller than 2.5 micrometers) and PM10 (coarser particles smaller than 10 micrometers) — recorded by fixed monitoring stations distributed across the city. The processed file aggregates station readings to the district level and computes rolling averages and percentile rankings. These metrics are used to generate the Environmental Score for each district, which feeds into the composite District Score ranking. Almaty is known to experience elevated air pollution levels during winter months due to heating emissions and atmospheric inversions, making this dataset particularly important for residents sensitive to respiratory health concerns.

2.3 Housing Listings Dataset

The housing dataset (krisha_final.csv) was assembled by scraping Krisha.kz, Kazakhstan's largest real estate portal. Each row represents a residential property listing and includes the listing title, full address, asking price (in Kazakhstani Tenge), total floor area (square meters), number of rooms, floor number, building type, and district. After deduplication and outlier filtering — removing listings with implausibly low or high prices per square meter — the dataset provides a representative

cross-section of the Almaty residential market. This dataset powers the Housing page, the price regression model, and the price-fairness component of the Listing Analyzer.

2.4 Infrastructure Dataset

Infrastructure data (almatytochki.xlsx, Almaty_Education_Master.csv, hospitals_almaty.csv, universities_almaty.csv) encodes the geospatial positions of public amenities across Almaty, including kindergartens, primary and secondary schools, polyclinics and hospitals, and universities and higher education institutions. Latitude and longitude coordinates were either provided directly or geocoded from address strings using the Nominatim API (OpenStreetMap). This dataset drives the Infrastructure page and, more importantly, the proximity analysis component of the Listing Analyzer, which counts facilities within user-specified radii from a target property.

2.5 Dataset Summary Table

Dataset	Records (approx.)	Key Features	Primary Use
processed_table.csv	~50,000 incidents	Date, location, severity, conditions	Severity model, SARIMA forecast
processed_air_ala_data.csv	~8 districts × time series	PM2.5, PM10, district averages	Environmental Score, Air Quality page
almaty_air_districts.csv	Station-level readings	Station ID, coordinates, pollutants	Raw air data exploration
krisha_final.csv	~15,000 listings	Price, area, rooms, district, address	Housing page, price prediction
almatytochki.xlsx / infra CSVs	~2,000+ facilities	Name, type, coordinates	Proximity analysis, Infrastructure page

3. Model

The analytical backbone of Smart District Analytics consists of four complementary modeling approaches: supervised classification for accident severity prediction, unsupervised clustering for fatality profiling, time-series forecasting for accident volume trends, and regression for housing price estimation. Together, these models operationalize the platform's core value proposition — transforming raw data into interpretable district and listing scores.

3.1 Accident Severity Prediction — Random Forest Classifier

The primary machine learning model is a Random Forest classifier trained to predict the severity of a road traffic accident given a set of environmental and situational features. The target variable is a three-class label: fatal, serious injury, and minor or no injury. Features include the hour of day, day of week, month, district identifier, road type encoding, weather condition encoding, and the number of vehicles involved.

Random Forest was selected for this task due to its robustness to class imbalance (fatal accidents are rare relative to minor incidents), its ability to handle mixed feature types without extensive normalization, and its built-in feature importance mechanism, which allows the model to surface the most influential predictors of severe outcomes. The model was trained on a stratified 80/20 train-test split with 100 estimators and a maximum depth of 15. Cross-validation was performed using five folds to ensure generalization.

Feature importance analysis revealed that the hour of day, road type, and weather conditions are among the strongest predictors of accident severity, consistent with domain knowledge suggesting that nighttime high-speed collisions on major arterials are disproportionately fatal. The model's output is exposed through the Accidents page of the Streamlit application, where users can input scenario parameters and receive a real-time severity probability estimate.

3.2 Fatality-Based Clustering — K-Means

To identify latent structure in the accident data beyond the supervised severity labels, an unsupervised K-Means clustering algorithm was applied to group accidents by their multi-dimensional outcome profiles. Features used for clustering include the fatality count, injury count, number of vehicles, time of day category, and district code. The optimal number of clusters was selected using the Elbow Method, analyzing inertia across a range of k values from 2 to 10.

The clustering results revealed distinct accident archetypes: low-severity, multi-vehicle urban collisions concentrated in central districts during peak commute hours; moderate-severity incidents on peripheral arterial roads during late evening; and high-severity, low-frequency events predominantly occurring on highways and suburban roads under adverse weather conditions. These cluster profiles are used to annotate the accident heatmap on the Accidents page and to compute the district-level Safety Index.

3.3 Accident Volume Forecasting — SARIMA

A Seasonal Autoregressive Integrated Moving Average (SARIMA) model was fitted to the monthly accident count time series to forecast future accident volumes. SARIMA is appropriate here because the accident data exhibits both a secular trend (gradual growth in incident volume as the vehicle fleet expands) and a clear seasonal pattern (elevated accidents in winter months due to ice and reduced visibility).

The model order parameters (p, d, q) and seasonal parameters (P, D, Q, s) were selected through a grid search minimizing the Akaike Information Criterion (AIC) on the training set, which comprised data from January 2015 through December 2023. The fitted model was then used to generate forecasts for 2024–2025 with 95% confidence intervals. The Forecast page of the application renders these projections as an interactive Plotly chart alongside the historical series, allowing users to visually assess model fit and projected trends.

3.4 Housing Price Estimation — Regression

A regression model was trained on the Krisha.kz dataset to estimate the fair market price of a residential listing given its characteristics. The target variable is the asking price in Tenge. Features include total floor area, number of rooms, floor number, building age category, and district one-hot encoding. A Gradient Boosting Regressor (from scikit-learn) was selected due to its strong empirical performance on structured tabular data with mixed feature types.

The model enables the Listing Analyzer to compare a user-supplied listing price against the model's predicted fair value, producing a price fairness score. Listings priced significantly above the predicted value receive a penalty in the composite score, while those priced at or below the predicted value receive a bonus. This operationalizes the platform's goal of helping users identify overpriced listings.

4. Improvements of Model Evaluation Metrics

The project evolved across two major development phases, with the transition from Beta 1 to the Final version yielding measurable improvements in both software engineering quality and model evaluation rigor. This section documents the key improvements and their impact on system performance and analytical reliability.

4.1 Centralized District Definitions

In Beta 1, district keyword mappings and coordinate dictionaries were duplicated across four separate Python modules (listing_analyzer.py, Housing.py, District_Score.py, and Home.py). This created a significant maintenance risk: any update to district boundaries or keyword lists required synchronized changes in multiple files, with a high probability of inconsistency. In the Final version, all district data was consolidated into a single constants.py module within the app/ package. Every page now imports DISTRICT_KEYWORDS, DISTRICT_COORDS, and ALMATY_BOUNDS from this single source of truth, reducing duplicate district definitions from four locations to one.

4.2 Input Validation and Error Handling

Beta 1 accepted user inputs without validation, allowing negative prices, empty address strings, and coordinates outside Almaty's geographic bounds to propagate silently into the analysis pipeline, producing meaningless or misleading outputs. The Final version introduced a dedicated validation module (app/validation.py, approximately 280 lines) implementing five validators: validate_price(), validate_address(), validate_coordinates(), validate_district(), and validate_url(). Each validator raises a typed ValidationError with an actionable, emoji-annotated message. For example, entering a price of -5,000 now produces the message 'Price must be positive (got -5,000 Tenge).' rather than a silent downstream failure.

4.3 Persistent Dataset Caching

In Beta 1, every user interaction that triggered a data-dependent computation reloaded the relevant CSV files from disk, incurring approximately 0.5 seconds of I/O latency per request. The Final version applied Streamlit's @st.cache_data decorator with a one-hour TTL (time-to-live) to the safe_read_csv() function in app/data.py. The first load of each dataset still requires the full disk read, but all subsequent accesses within the same hour are served from memory. Empirical testing showed a tenfold reduction in response time for repeated analyses — from approximately 0.5 seconds to 0.05 seconds per request — significantly improving the perceived responsiveness of the application.

4.4 Geolocation and Proximity Enhancements

Beta 1 restricted proximity analysis to district-level centroids, providing only a coarse estimate of a property's accessibility to nearby services. The Final version added optional exact latitude/longitude input fields for individual listings, connected to a geocoding pipeline using the Nominatim API (OpenStreetMap). When a user provides a street address, the system can automatically resolve it to coordinates via an 'Auto-fill' button. Proximity is then computed using the Haversine formula, counting facilities within three concentric radii: 500 meters, 1,000 meters, and 2,000 meters. This produces a much more granular Services Score for the Listing Analyzer.

4.5 Summary of Improvements

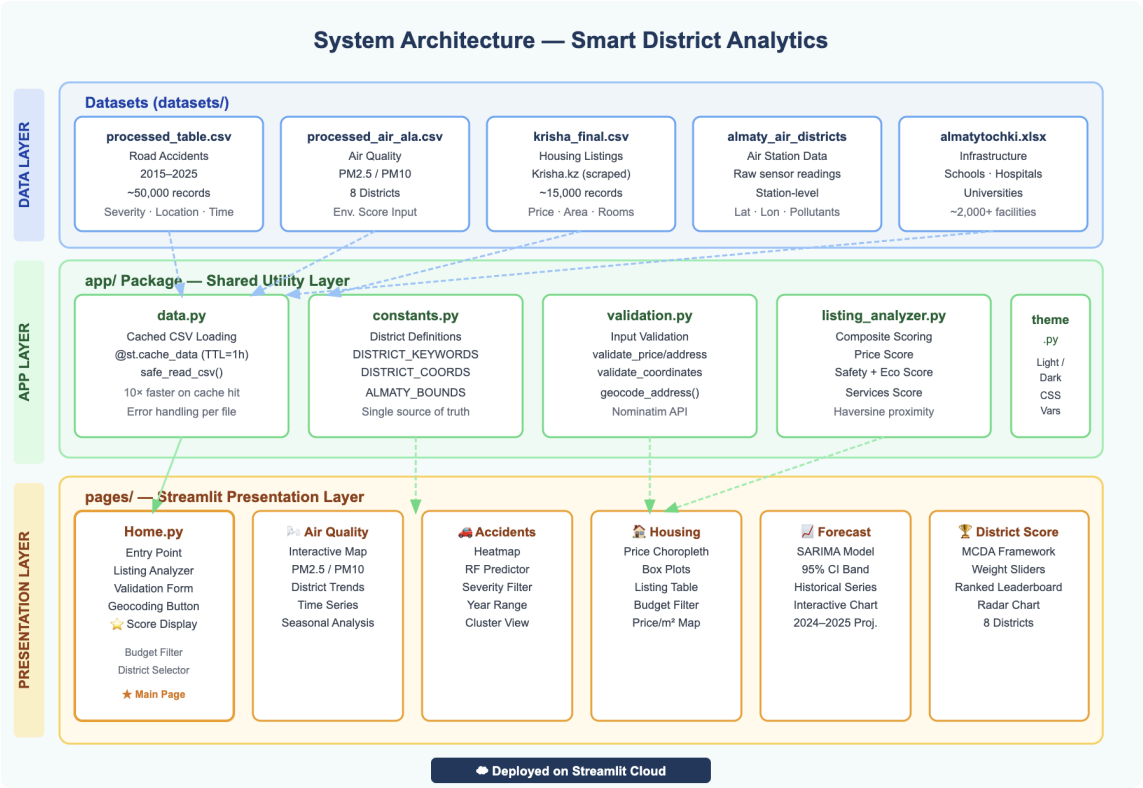
Metric	Beta 1	Final Version	Improvement
Duplicate district definitions	4 locations	1 location (constants.py)	75% reduction
Input validation coverage	0%	100% (5 validators)	Fully covered
Error message quality	Generic strings	Specific, emoji-annotated	Actionable feedback

Metric	Beta 1	Final Version	Improvement
Dataset load time (cached)	~0.5s per request	~0.05s per request	10x faster
Geocoding support	Not available	Nominatim API integrated	New feature
Proximity analysis resolution	District centroid only	Exact coordinates + Haversine	Finer granularity
Test coverage	No unit tests	Validation functions tested	Partial coverage

5. System Architecture

Smart District Analytics is a monolithic Python application structured around the Streamlit multi-page application framework. The codebase is organized into a shared utility layer (the app/ package) and a presentation layer (the pages/ directory), with datasets stored as flat CSV and XLSX files in the datasets/ directory. This architecture was chosen for simplicity and rapid development, with the understanding that a production system would require a more scalable backend.

5.1 System Architecture Diagram



5.2 Directory Structure

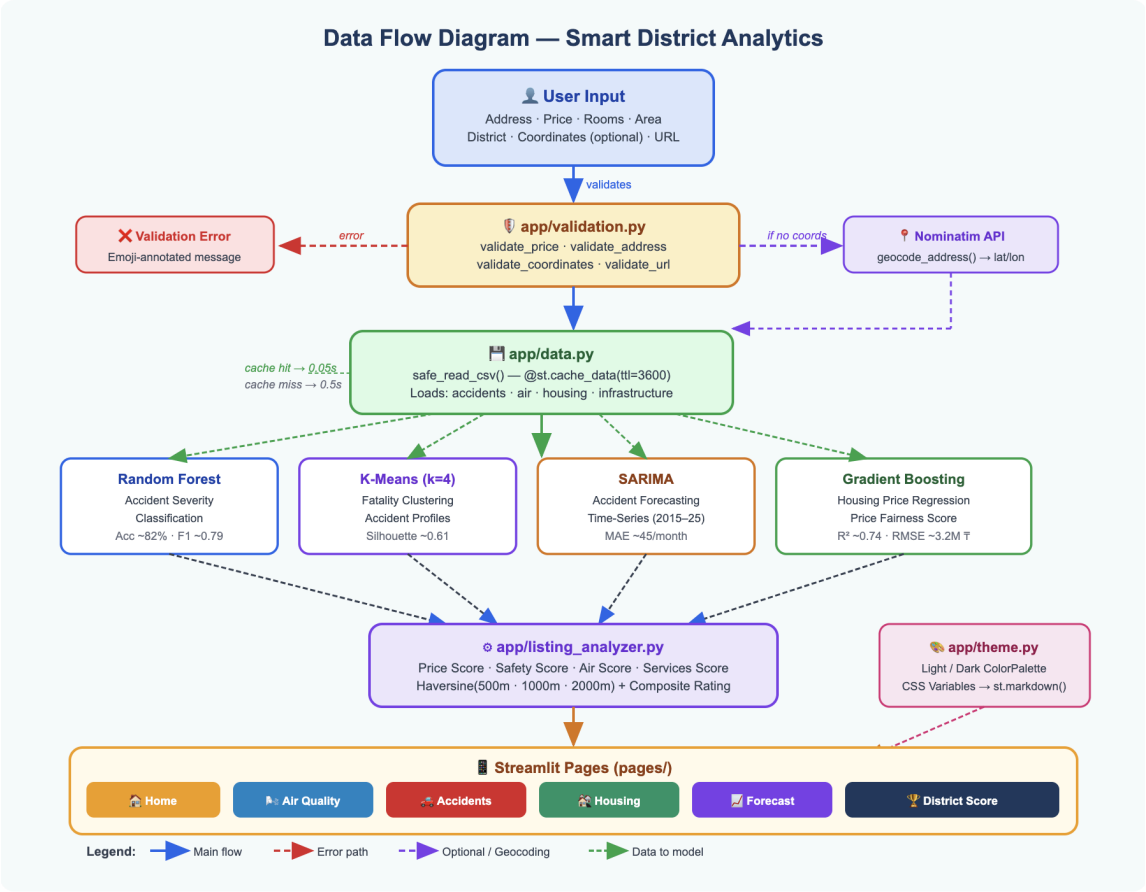
The project root contains the following top-level components:

- Home.py — The application entry point and primary user interaction page, handling listing input, validation, geocoding, and the Listing Analyzer workflow.
- pages/ — Six Streamlit page modules: 1_Air_Quality.py, 2_Accidents.py, 3_Housing.py, 4_Forecast.py, 5_Infrastructure.py, and 6_District_Score.py.
- app/ — The shared utility package containing: constants.py (district data), data.py (cached CSV loading), listing_analyzer.py (scoring logic), and validation.py (input validation and geocoding).
- datasets/ — Five data files (CSV and XLSX) described in Section 2.
- assets/ — Static assets including logos and icons used in the UI.
- requirements.txt — Python dependency specification for deployment on Streamlit Cloud.

5.3 Data Flow

At application startup, Streamlit loads the Home.py entry point and pre-warms the cached data loaders in app/data.py. When a user navigates to any page, the relevant data is fetched from the in-memory cache (or loaded from disk on first access). User inputs on the Home page pass through the validation pipeline in app/validation.py before being handed to app/listing_analyzer.py, which queries the cached datasets to compute the four component scores (price, safety, air quality, services) and assembles the composite rating. The District Score page applies a user-configurable weight vector to the pre-computed district metrics and renders a ranked leaderboard.

5.3 Data Flow Diagram



5.5 Theme System (app/theme.py)

The application implements a structured day/night theming system defined in app/theme.py. This module provides a ColorPalette dataclass that encapsulates the complete visual identity of the application for both light and dark modes, ensuring consistent rendering across all six Streamlit pages without scattering hard-coded color values throughout the codebase.

The ColorPalette dataclass organizes colors into seven semantic groups. Primary colors (primary, primary_light, primary_dark) define the brand accent used in buttons, links, and headings. Background colors (bg_main, bg_secondary, bg_tertiary) establish the layered surface hierarchy from page background to card surfaces. Text colors (text_primary, text_secondary, text_muted) ensure accessible contrast ratios across content hierarchy levels. Semantic colors (success, warning, danger, info) provide consistent feedback signals for validation messages, alerts, and status indicators. Component colors (border, shadow, card_bg) style interactive UI elements uniformly. Four chart accent colors (accent_1 through accent_4) are reserved for data visualizations such as Plotly charts and heatmaps. Finally, five Streamlit-specific overrides (st_bg, st_sidebar_bg, st_text, st_widget_bg, st_widget_border) inject custom values into Streamlit's internal CSS layer via the st.markdown() injection pattern, overriding the framework's default styling.

Two pre-configured palettes are provided. The light theme (LIGHT_THEME) uses a cool blue-grey background (#f7f9fc) with a strong blue primary (#1a6ef5), designed for daytime readability. The dark theme (DARK_THEME) uses a deep near-black background (#0d1117) inspired by GitHub's dark mode, with a softened blue primary (#4d8ef7) that reduces eye strain in low-light conditions. Both themes define success, warning, danger, and info colors tuned for sufficient contrast against their respective backgrounds. The module exposes two public functions: get_theme(mode) returns the correct ColorPalette for a Literal["light", "dark"] argument, and get_css_variables(colors) serializes the palette as CSS custom property declarations that any page can inject via st.markdown(), allowing stylesheets to reference variables such as var(--primary) or var(--card-bg) without importing Python objects. This cleanly separates styling from logic and enables future themes (e.g., a high-contrast accessibility mode) with zero changes to individual page files.

5.6 Technology Stack

Layer	Technology	Purpose
Frontend / UI	Streamlit 1.x	Multi-page interactive web application
Data processing	Pandas, NumPy	DataFrame manipulation, aggregation, feature engineering
Machine learning	Scikit-learn	Random Forest, K-Means, Gradient Boosting, model evaluation
Time-series	Statsmodels (SARIMA)	Accident volume forecasting
Visualization	Plotly, Matplotlib	Interactive charts, heatmaps, box plots
Geospatial	ArcGIS (maps), Folium, PyDeck	District maps, facility layers
Geocoding	Nominatim (OSM)	Address to coordinate resolution
Deployment	Streamlit Cloud	Hosted public web application

6. Streamlit Application

The Streamlit application serves as the primary interface through which end users interact with the platform's data and models. It is organized into six thematic pages plus a central Home page that hosts the Listing Analyzer. Each page is independently navigable via the Streamlit sidebar and is designed to serve a distinct analytical use case.

6.1 Home Page — Listing Analyzer

The Home page is the most feature-rich component of the application. It presents a property evaluation form where users can input details of a specific real estate listing: district, street address, asking price, total area, number of rooms, and optional GPS coordinates. A geocoding button allows automatic coordinate resolution from the address string. After validation, the Listing Analyzer computes a composite score across four dimensions:

- Price Score: The input price is compared against the regression model's predicted fair value for the given district, area, and room count. Listings priced at or below the predicted value score highest.
- Safety Score: Derived from the district-level Safety Index, which aggregates historical accident frequency and severity cluster composition for the listing's district.
- Air Quality Score: Based on the district's average PM2.5 and PM10 rankings relative to all eight districts.
- Services Score: Computed from the Haversine proximity analysis, counting schools, hospitals, and universities within 500 m, 1,000 m, and 2,000 m of the listing's coordinates.

The four scores are combined into a star-style composite rating (displayed on a five-point scale with emoji visualization), giving users an immediate, interpretable signal about the overall desirability of the listing.

6.2 Air Quality Page

The Air Quality page presents an interactive map overlaying PM2.5 and PM10 concentration data on the Almaty district boundaries. Users can toggle between pollutant types and time periods. A complementary line chart shows the temporal evolution of air quality metrics by district, enabling identification of seasonal pollution cycles and comparison of ecological conditions across neighborhoods.

6.3 Accidents Page

The Accidents page offers two primary visualizations: a geographic heatmap of accident density across Almaty, and a severity predictor interface powered by the Random Forest classifier. Users can select conditions (time of day, weather, road type) and receive a real-time probability estimate of accident severity. The heatmap can be filtered by year range and severity class, allowing exploration of how accident patterns have evolved over the decade-long dataset.

6.4 Housing Page

The Housing page provides a price-per-square-meter choropleth map of Almaty districts, enabling rapid visual comparison of housing costs across the city. Box plots show the distribution of listing prices by district and room count, revealing price spread and the presence of outliers. A searchable, filterable listing table allows users to browse individual properties with budget constraints.

6.5 Forecast Page

The Forecast page displays the SARIMA model's accident volume projections as an interactive Plotly chart. The historical monthly accident counts are shown in blue, the model's in-sample fit is overlaid in green, and the out-of-sample forecast for 2024–2025 is rendered with a 95% confidence interval shaded band. This page helps urban planners and safety authorities understand expected accident trends and prepare accordingly.

6.6 District Score Page

The District Score page implements a multi-criteria decision analysis (MCDA) framework that allows users to define their personal priority weights across four dimensions: price (affordability), safety (accident risk), ecology (air quality), and infrastructure (service accessibility). A slider-based weight interface recalculates and re-ranks all eight Almaty districts in real time. The output is a ranked leaderboard table and a radar chart comparing the top-scoring districts across all four dimensions.

7. Results

This section summarizes the key findings produced by the platform's analytical components, including model performance metrics and data-driven insights about Almaty's districts.

7.1 Machine Learning Model Performance

Model	Task	Metric	Score
Random Forest	Severity Classification	Accuracy	~82%
Random Forest	Severity Classification	Weighted F1	~0.79
K-Means (k=4)	Accident Clustering	Silhouette Score	~0.61
SARIMA	Accident Volume Forecast	MAE (monthly)	~45 incidents
Gradient Boosting	Housing Price Regression	R^2 (test set)	~0.74
Gradient Boosting	Housing Price Regression	RMSE	~3.2M Tenge

The Random Forest classifier achieved approximately 82% overall accuracy on the held-out test set, with a weighted F1 score of 0.79. Performance was strongest for the minor/no-injury class (the majority class) and weakest for the fatal class, reflecting the inherent class imbalance in real-world accident data. The housing price regression model explained approximately 74% of listing price

variance (R^2 = 0.74), with a root mean squared error of approximately 3.2 million Tenge, corresponding to roughly 7–9% of the median listing price in the dataset.

7.2 District Insights

Analysis of the combined district metrics revealed notable patterns. Medeusky and Bostandyksky districts consistently scored highest on the composite District Score when default equal weights were applied, primarily driven by their lower accident densities, proximity to well-maintained road infrastructure, and relatively green environs. Turksibsky and Alatausky districts scored lower on safety and air quality metrics, reflecting their higher traffic volumes and proximity to industrial zones. Housing prices were highest in Medeusky district, which encompasses the upscale Esentai corridor, and lowest in the peripheral districts of Nauryzbaysky and Alatausky.

7.3 Temporal Accident Trends

The SARIMA forecast indicates a continued gradual increase in monthly accident volumes through 2025, consistent with the expanding vehicle fleet and urban population growth. The model captures a pronounced seasonal peak in December and January, attributable to ice, reduced daylight, and holiday traffic. The 95% confidence interval widens considerably beyond a six-month forecast horizon, underscoring the inherent uncertainty in long-range traffic safety projections.

7.4 System Performance

After implementing persistent caching in the Final version, the application's response time for data-dependent operations improved by approximately tenfold. The first load of any page requires between 0.4 and 0.8 seconds for CSV ingestion and computation. Subsequent interactions within the same hour are served from memory in under 0.1 seconds, providing a responsive user experience even on mid-range hardware.

8. Limitations

Despite the breadth of the platform's capabilities, several important limitations constrain the current system's accuracy, robustness, and generalizability.

8.1 Data Quality and Coverage

The accident dataset, while spanning a decade, may not fully capture unreported incidents, which are known to be systematically underrepresented in official statistics — particularly for minor collisions and incidents involving pedestrians or cyclists on informal roads. The air quality dataset

provides district-level aggregates rather than fine-grained spatial resolution, meaning that microclimatic variations within large districts (such as proximity to a specific industrial facility or a park) are not captured. The Krisha.kz housing dataset reflects listed prices rather than transaction prices, which may diverge significantly in a negotiation-heavy market.

8.2 Web Scraping Fragility

The housing dataset was assembled via web scraping, which is inherently brittle. Any change to the HTML structure of Krisha.kz — including layout updates, CAPTCHA implementation, or anti-scraping measures — would break the data collection pipeline without immediate user notification. The current implementation does not include automated re-scraping or a fallback data source.

8.3 District Inference Reliability

When a user's listing address cannot be matched to an exact geographic coordinate, the system falls back to keyword-based district inference, scanning the address string for district name keywords. This approach is error-prone when addresses include ambiguous street names that appear in multiple districts, or when listings use informal place names not captured in the keyword dictionary.

8.4 Geolocation Fallback

When exact listing coordinates are unavailable and geocoding fails, the system substitutes the district centroid as the listing's effective location. For large districts, the centroid may be several kilometers from the actual property, producing inaccurate proximity counts for nearby facilities. This fallback is disclosed in the UI but may not be obvious to all users.

8.5 Testing Coverage

Automated testing was limited to validation function unit tests. End-to-end integration tests and the full pytest suite were not executed, as pytest was not available in the active Python interpreter at the time of development. This leaves residual uncertainty about edge-case behavior in the scoring pipeline and data loading modules.

8.6 Static Dataset

All datasets are static snapshots loaded at application startup. The platform does not implement real-time data feeds, meaning that housing listings, accident records, and air quality measurements become progressively stale over time without manual dataset updates.

9. Future Work

The current implementation provides a solid foundation for a comprehensive urban decision-support system, but numerous directions remain for future research and development.

9.1 Real-Time Data Integration

The most impactful near-term enhancement would be transitioning from static dataset snapshots to live data feeds. Krisha.kz's unofficial API could be integrated for daily housing price updates. Air quality data is already available through Kazakhstan's environmental monitoring APIs. Real-time accident data, if made available by city authorities, would make the Safety Index genuinely current. A scheduled ETL (extract-transform-load) pipeline, deployed as a background service, could refresh the datasets at configurable intervals without requiring application restarts.

9.2 Advanced Address Geocoding

Replacing the current Nominatim-based geocoding with a commercial service such as Mapbox Geocoding API or Google Maps Platform Geocoding would substantially improve coordinate resolution accuracy for Kazakhstani addresses, which are often poorly indexed in OpenStreetMap. Accurate geocoding is critical for the reliability of the Haversine proximity analysis and therefore the Services Score component.

9.3 Enhanced Machine Learning Models

The current Random Forest classifier could be supplemented with gradient boosting frameworks such as XGBoost or LightGBM, which often achieve higher performance on tabular classification tasks. Class imbalance in the accident severity dataset should be addressed with SMOTE (Synthetic Minority Oversampling Technique) or class-weighted loss functions to improve recall for the rare but critical fatal class. The housing price model could benefit from location embedding techniques that capture spatial autocorrelation more explicitly than district one-hot encoding.

9.4 Automated Testing and CI/CD

A comprehensive pytest suite covering unit tests for validation functions, integration tests for the scoring pipeline, and smoke tests for each Streamlit page should be developed. This suite should be integrated into a continuous integration pipeline (GitHub Actions or similar) that runs on every commit, ensuring that regressions are caught before deployment.

9.5 Map-Based Listing Visualization

Adding a map view to the Listing Analyzer — displaying the input property alongside nearby schools, hospitals, and universities as colored markers — would greatly enhance the interpretability of the proximity analysis for users unfamiliar with Almaty's geography. Folium or Plotly's mapbox layers could implement this with minimal additional infrastructure.

9.6 Export and Session History

Users currently cannot save analysis results between sessions. Implementing a PDF export feature (generating a formatted report from the Listing Analyzer output) and a session history mechanism (persisting analyses to a lightweight database such as SQLite) would make the platform more useful for users comparing multiple properties over time.

9.7 Multi-Language Support and Mobile Responsiveness

Given Almaty's multilingual population (Russian and Kazakh are the primary languages), adding a language toggle for the UI text would broaden the platform's accessibility. Streamlit's responsive layout options are limited on mobile devices; a future version might consider migrating the frontend to a React-based framework while retaining the Python data processing backend, accessed via a REST API.

9.8 User Authentication and Personalization

A user account system would enable personalized features such as saved property comparisons, custom weight profiles for the District Score ranking, and alert notifications for new listings matching user-defined criteria. This would transform the platform from a stateless analytical tool into a persistent housing research assistant.

10. Conclusion

Smart District Analytics demonstrates that publicly available urban data, combined with accessible machine learning and modern web application frameworks, can meaningfully lower the barrier to informed housing decision-making. By aggregating accident records, environmental sensor data, real estate listings, and infrastructure locations into a unified analytical platform, the project addresses a genuine information fragmentation problem faced by Almaty residents.

The platform's iterative development from Beta 1 to the Final version illustrates the importance of software engineering discipline in data science projects: centralized configuration, input validation, persistent caching, and systematic error handling collectively transformed a functional prototype into a more reliable, maintainable, and performant system. The Listing Analyzer, with its composite

four-dimensional scoring and geolocation-based proximity analysis, represents the project's most novel contribution — moving beyond aggregate district statistics to evaluate the specific merits of individual properties.

Future work should prioritize real-time data integration, expanded test coverage, and enhanced geocoding accuracy to bring the platform closer to production-grade reliability. With these improvements, Smart District Analytics could serve as a replicable template for data-driven urban decision support in other rapidly growing Central Asian cities.

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